

ABDURAHMAN ALARCON

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EDUCATION

Northern Virginia Community College
A.S. in **Information Technology**

Graduated 2025

GEORGE MASON UNIVERSITY - College of Engineering & Computing
B.S. in **Information Technology** – Concentration in **Cybersecurity**

Expected May 2027

SKILLS

Proficient Languages: Python, UML, Java, JavaScript, Bash, SQL, PowerShell

Technical: Systems Management – ServiceNow, JAMF, Intune, SCCM, Active Directory, Tenable, Cisco networking

AI/ML – Sentence-Transformers, FAISS, LangChain, RAG pipelines, MITRE ATT&CK framework

Cloud & Tools – AWS, Azure, Node.js, MS Office Suite (Excel, Word, PowerPoint), Cameo, UML

Networking – TCP/IP, DNS, DHCP, MPLS, VLANs, Firewall Policy, AWS VPC, Wireshark

EXPERIENCE

LIBRARY OF CONGRESS

Washington, DC

Network Engineering Intern

May 2026 – Present

- Configured and troubleshot TCP/IP, DNS, DHCP, and MPLS across enterprise environments, supporting VLAN segmentation and firewall policy management to enforce traffic control and network security.
- Monitored and supported AWS VPC infrastructure including subnet routing and security group configuration, ensuring cloud connectivity and access control across the organization.
- Automated endpoint provisioning via PowerShell and Bash scripts across Windows and macOS, and managed device identity through Active Directory, Intune, and ServiceNow — resolving ~20 network-related tickets per week.

COMMONWEALTH CYBER INITIATIVE

Fairfax, VA

Undergraduate Cybersecurity Researcher

Feb 2026 – Present

- Built an AI inference model using MiniLM to map non-KEV CVEs lacking ATT&CK entries to MITRE ATT&CK techniques, expanding coverage to 2,000+ CVEs
- optimized lightweight transformer-based NLP workflows, using FAISS for similarity search, to match vulnerability descriptions with adversarial tactics and techniques across cybersecurity datasets.
- Presented progress and findings to NSWC Dahlgren stakeholders to align research direction with evolving DoD-relevant threat assessment goals.

OUTAMATION

Fairfax, VA

AI Machine Learning Engineer Intern

July 2025 – Aug 2025

- Optimized mortgage document processing using Python and AI models to classify, standardize and extract salaries, dates, account numbers, and other key data fields leading to increased efficiency.
- Performed analysis and testing of AI models (DocLLM, LangChain, LayoutLM) for file classification and data extraction to identify optimal tools for mortgage industry applications to the customer.
- Delivered actionable plans and analysis results for optimizing document processing workflows and leveraged benchmarked AI tools to enhance data accuracy and operational efficiency.

DEPARTMENT OF STATE

Fairfax, VA

Software Engineer Intern

May 2024 – Aug 2024

- Deployed an Azure-based mobile application backend using Azure SQL and Node.js, improving access to critical resources for 200 U.S. expatriate users and reducing inquiry calls by 20%.
- Automated IT network security protocols through PowerShell scripting and systematically debugged software and conducted maintenance which reduced DOS employee service request tickets by 30%.
- Maintained Cisco networking systems and Linux servers across embassy departments, ensuring 100% system uptime and secured operations for 3 critical services for interdepartment communication.

TECHNICAL PROJECTS

MORTGAGE RAG PIPELINE (AI Data Processing)

2025

- Engineered a Retrieval Augmented Generation (RAG) pipeline to enhance LLM accuracy; developed a data ingestion pipeline to chunk and embed over 1,000 documents, reducing model hallucination by 25%.
- Optimized retrieval for a RAG pipeline by implementing hybrid search and re-ranking algorithms which decreased query latency by 40%.
- Managed performance and maintenance of a RAG pipeline by creating a continuous evaluation framework, ensuring long-term reliability by proactively addressing data drift.